

# PAPER\_483 — Multi-System UQFF Compiler Modules: 4/5/7/8 AstroSystems Architecture

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## Abstract

This paper documents the **multi-system compiler modules** extracted from `grok_share_bdfb3a05b` — a new architectural tier above individual-system modules. Where PAPER\_481 documents single-system UQFF implementations, these modules simultaneously compute  $F_{U_{Bi_i}}$  for 4, 5, 7, and 8 astrophysical systems through a single `computeMasterEquations(system)` call, using a runtime-dispatched parameter lookup (`setSystemParams`). New physics includes the triadic simultaneous solver (`computeTriadicSolution`), gas-nebula integration (`computeGasNebulaIntegration`), and the Source160 computation validation showing  $F_{U_{Bi_i}} \approx -8.32 \times 10^{217}$  N universally for template-mass systems.

**Source:** `grok_share_bdfb3a05b06.txt`, docx: “4 Astro Systems\_11Oct2025”, “5 Astro Systems\_11Oct2025”, “7 Astro Systems\_11Oct2025”, “8 Astro Systems\_11Oct2025”, “8 Astro Systems\_B\_11Oct2025”, “Source160.docx”. Analyzed Oct 23, 2025.

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## 1. Module Hierarchy

### UQFF Multi-System Architecture

- AstroSystemsUQFFModule (4 systems)  
Systems: NGC685, NGC3507, NGC3511, AT2024tvd  
New API: `computeMasterEquations(system)`, `setSystemParams(system)`  
`computeGravityCompressed(system)`, `computeResonanceUr(U_dp, U_r, system)`  
`computeBuoyancyUbi(system)`
- UQFFNebulaTriadicModule (5→7 systems, 2 canonical versions)  
Systems v5: NGC3596, NGC1961, NGC5335, NGC2014 + Carina  
Systems v7: NGC685, NGC3507, NGC3511, Carina + 3 extended  
New API: `computeGasNebulaIntegration(system)` ← unique nebular term
- UQFFBuoyancyModule 7-system (multi-system buoyancy variant)  
Systems: M74, M16, M84, Centaurus A (4+ systems)  
Source: “7 Astro Systems\_11Oct2025”
- UQFF8AstroSystemsModule (8 systems)  
Systems: NGC4826, NGC1805, NGC6307, NGC7027 + 4 more  
New API: full 8-system dispatch via `computeMasterEquations(system)`
- UQFF8AstroTriadicModule (8 systems, triadic solver)  
Systems: Same 8 as above  
New API: `computeTriadicSolution(system, t)` ← simultaneous triadic solve  
`computeGravityCompressed(system)`, `computeResonanceUr(U_dp, U_r, system)`  
`computeBuoyancyUbi(system)`

## 2. System-Specific Parameter Registries

### 2.1 AstroSystemsUQFFModule (4 Systems)

**Comment doc:** "Multi-system params: NGC685 M=1e41 kg r=1e21 m; NGC3507 M=2e41 kg r=2e21 m; NGC3511 M=3e41 kg r=3e21 m; AT2024tvd M=1..."

| System    | M (kg)               | r (m)              | Type                                |
|-----------|----------------------|--------------------|-------------------------------------|
| NGC685    | $10^{41}$            | $10^{21}$          | Barred spiral galaxy                |
| NGC3507   | $2 \times 10^{41}$   | $2 \times 10^{21}$ | Barred lenticular galaxy            |
| NGC3511   | $3 \times 10^{41}$   | $3 \times 10^{21}$ | Spiral galaxy                       |
| AT2024tvd | $10^{41}$ (template) | $10^{21}$          | Sept 2025 transient TDE off-nucleus |

**AT2024tvd** is notable — this is the off-nucleus tidal disruption event discovered September 2024 in a galaxy 600 Mpc away, consistent with UQFF cluster-scale parameters.

### 2.2 UQFFNebulaTriadicModule (5 Systems)

**Comment doc:** "Multi-system params: NGC3596 M=1e41 kg r=1e21 m; NGC1961 M=2e41 kg r=2e21 m; NGC5335 M=3e41 kg r=3e21 m; NGC2014 M=4e..."

| System  | M (kg)             | r (m)                   | Type                              |
|---------|--------------------|-------------------------|-----------------------------------|
| NGC3596 | $10^{41}$          | $10^{21}$               | Spiral galaxy (Leo cluster)       |
| NGC1961 | $2 \times 10^{41}$ | $2 \times 10^{21}$      | Peculiar spiral (AGN)             |
| NGC5335 | $3 \times 10^{41}$ | $3 \times 10^{21}$      | Lenticular galaxy                 |
| NGC2014 | $4 \times 10^{41}$ | $\sim 4 \times 10^{21}$ | LMC H II region                   |
| Carina  | $\sim 10^{36}$     | $\sim 10^{17}$          | Carina Nebula star-forming region |

### 2.3 UQFFNebulaTriadicModule (7 Systems — Canonical Version)

**Comment doc:** "Multi-system params: NGC685 M=1e41 kg r=1e21 m; NGC3507 M=2e41 kg r=2e21 m; NGC3511 M=3e41 kg r=3e21 m; Carina M=1e36..." - 4 systems from 4AstroSystems + Carina + 2 additional (from 7-system batch)

### 2.4 UQFF8AstroSystemsModule (8 Systems)

**Comment doc:** "Multi-system params: NGC4826 M=1e41 kg r=1e21 m; NGC1805 M=2e41 kg r=2e21 m; NGC6307 M=3e41 kg r=3e21 m; NGC7027 M=..."

| System  | M (kg)             | r (m)              | Type                |
|---------|--------------------|--------------------|---------------------|
| NGC4826 | $10^{41}$          | $10^{21}$          | Black Eye Galaxy    |
| NGC1805 | $2 \times 10^{41}$ | $2 \times 10^{21}$ | Open cluster in LMC |

| System  | M (kg)             | r (m)              | Type                           |
|---------|--------------------|--------------------|--------------------------------|
| NGC6307 | $3 \times 10^{41}$ | $3 \times 10^{21}$ | Elliptical galaxy              |
| NGC7027 | —                  | —                  | Planetary nebula (JWST 2022)   |
| +4 more | —                  | —                  | From “8 Astro Systems_B” batch |

### 3. New API Methods

#### 3.1 computeMasterEquations(system) — Multi-System Dispatcher

```
cdouble AstroSystemsUQFFModule::computeMasterEquations(const std::string& system) {
    setSystemParams(system); // Populate variables["M"], variables["r"] for chosen sys
    return computeF(t_default); // Full F_U_Bi_i for that system's parameters
}
```

Supports runtime system switching:

```
AstroSystemsUQFFModule mod;
auto F_NGC685 = mod.computeMasterEquations("NGC685");
auto F_NGC3507 = mod.computeMasterEquations("NGC3507");
auto F_AT2024 = mod.computeMasterEquations("AT2024tvd");
```

#### 3.2 computeGasNebulaIntegration(system) — NEW (UQFFNebulaTriadicModule only)

Unique to the nebular triadic module — computes additional gas-phase integration term:

$$F_{gas} = \int \rho_{gas}(r) \cdot g_{UQFF}(r, t) dV \approx \rho_{gas} \cdot g_{UQFF} \cdot V_{nebula}$$

This term accounts for the ISM/gas column density contribution to the UQFF total field in nebular environments, scaling with  $L_{H\alpha}$  and SFR. Specific to NGC3596, NGC1961, NGC5335, NGC2014, and Carina Nebula — all star-forming or nebular gas systems.

#### 3.3 computeResonanceUr(U\_dp, U\_r, system) — Resonance Mode Selector

```
cdouble AstroSystemsUQFFModule::computeResonanceUr(int U_dp, int U_r, const std::string& system) {
    // U_dp = DPM resonance mode (0=off, 1=on)
    // U_r = resonant coupling mode (0=magnetic, 1=vacuum)
    // system = dispatch key
}
```

Extends DPM resonance with two independent control flags for fitting observational constraints.

#### 3.4 computeTriadicSolution(system, t) — NEW (UQFF8AstroTriadicModule only)

Simultaneously solves all three triadic arms for a given system and time:

$$\vec{F}_{triadic} = (F_{compressed}(t), F_{resonant}(t), F_{buoyancy}(t))$$

Returns the triadic solution vector as three cdouble values, enabling simultaneous constraint by compressed, resonant, and buoyancy observational data.

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## 4. Source160 Validation Data (Thoughts L10436)

From the Grok Thoughts section analyzing Source160.docx:

### 4.1 Full $F_{U_{Bi_i}}$ (template-mass systems)

$$F_{U_{Bi_i}} \approx -8.32 \times 10^{217} + i(-6.75 \times 10^{160}) \text{ N}$$

Identical for: **Tycho's SNR, Abell 2256, Tarantula Nebula, NGC 253** — when evaluated at template-mass ( $M = 10^{41}$  kg,  $r = 10^{21}$  m,  $\omega_0 = 10^{-15}$ ).

### 4.2 Compressed Integrand per System

$$F_{U_{Bi_i}, integrand} = \sum \text{terms} \approx \begin{cases} 6.16 \times 10^{45} \text{ N} & \text{Abell 2256 } (M = 1.23 \times 10^{45} \text{ kg}) \\ 6.16 \times 10^{39} \text{ N} & \text{Template mass systems} \end{cases}$$

### 4.3 DPM Resonance per System

$$\text{DPM}_{res} = \frac{g_L \mu_B B_0}{\hbar \omega_0} \approx \begin{cases} 1.76 \times 10^{15} & \text{Tycho's SNR} \\ 1.76 \times 10^{17} & \text{Abell 2256} \\ 1.76 \times 10^{18} & \text{Tarantula, NGC 253} \end{cases}$$

### 4.4 Buoyancy $U_{b1}$ (all systems universal)

$$U_{b1} = \beta_i V_{infl,UA} \rho_{vac,A} a_{univ} \approx (6 \times 10^{-19} + 6.6 \times 10^{-20}i) \text{ N}$$

### 4.5 Superconductive $U_i$ (all systems, $t = t_0$ )

$$U_i \approx (1.7 \times 10^{-43} + 1.2 \times 10^{-44}i) \text{ N}$$

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## 5. Architecture Notes

### 5.1 File Inventory (Created Session 126)

| File                         | Size (chars) | Notes                                                              |
|------------------------------|--------------|--------------------------------------------------------------------|
| AstroSystemsUQFFMod31587     | 31,587       | 4-system dispatcher w/ setSystemParams, computeGravity-Compressed  |
| AstroSystemsUQFFMod14941p    | 14,941p      |                                                                    |
| UQFFNebulaTriadicMod34720h   | 34,720h      | 5+7-system, compute-GasNebulaIntegration canonical largest version |
| UQFFNebulaTriadicMod151343pp | 151,343pp    |                                                                    |
| UQFF8AstroSystemsMod41144h   | 41,144h      | 8-system dispatcher largest 8-system implementation                |
| UQFF8AstroSystemsMod171211pp | 171,211pp    |                                                                    |
| UQFF8AstroTriadicMod41165h   | 41,165h      | 8-system triadic solver                                            |

| File                        | Size (chars) | Notes                    |
|-----------------------------|--------------|--------------------------|
| UQFF8AstroTriadicModule.cpp | 3,681        | computeTriadicSolution() |

## 5.2 Deduplication Status

- UQFFNebulaTriadicModule: appeared at L9597 (5-system) and L9990 (7-system/canonical) — L9990 version is canonical (larger, 320 cpp lines vs 289)
- UQFF8AstroSystemsModule: appeared at L10539, L10672, L11310, L11429, L11541 — L10539 version is canonical (383 cpp lines, largest)
- UQFFBuoyancyModule (7-system at L10102): 368-line cpp variant — skipped (UQFF-BuoyancyModule.h/.cpp already exist from PAPER\_479)

## 6. CP2 + CP4 Integration

### Phase C (Session 126): CP2 Additions

Add three calculators to CondensedPhysics2.py: 1. UQFFBuoyancyAstroCalculator — Session 125 pending (CP2: 600→601) 2. UQFFBuoyancyCNBCalculator — Session 125 pending (CP2: 601→602) 3. IndividualSystemUQFF18Calculator — Session 126 (CP2: 602→603) 4. MultiSystemUQFFCompilerCalculator — Session 126 (CP2: 603→604) 5. HydrogenResonanceUQFFCalculator — Session 126 (CP2: 604→605)

### Phase D (Session 125+126): CP4 Additions

- Session125GrokShare4e4d8be1f7HubCalculator (#104) — pending from Session 125
- Session126GrokShareBdfb3a05b06HubCalculator (#105) — this session

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                   | UQFF Prediction                                                                     | SM / Experiment                                | Source                                          | Alignment                                                    |
|------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------|
| Higgs mass<br>$m_H$          | UQFF<br>$K_{\text{HIGGS}}=47.34 \rightarrow$<br>$m_H_{\text{UQFF}} =$<br>125.09 GeV | $m_H = 125.20 \pm$<br>0.11 GeV                 | PDG<br>2024                                     | 99.8%                                                        |
| Cosmological<br>$\Lambda$    | UQFF                                                                                | $\nabla \text{UA}$                             | $^2 \rightarrow$<br>1.09e-52<br>$\text{m}^{-2}$ | $\Lambda =$<br>1.114e-52<br>$\text{m}^{-2}$<br>(Planck+DESI) |
| Thomson $\sigma_T$<br>(QED)  | UQFF $U_m$ kernel:<br>$\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$                | $\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$ | PDG<br>2024                                     | 100% (exact)                                                 |
| $\kappa$ baryon<br>stability | $\kappa = 0.0005/\text{day};$<br>scale separation<br>$10^{33}$ from proton<br>decay | $\tau_p > 7.7\text{e}33$ yr<br>(Super-K)       | Super-K<br>2024                                 | ✓ UQFF<br>baryon-safe                                        |

**New physics claim:** UQFF operates at a vacuum topology scale ( $\sim 200$  PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite *PAPER\_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

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